Project: Worked examples
Title: Shaft under combined bending and torsion
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Bending and torsional stresses in a solid round shaft with stress-concentration factors, the maximum shear (Tresca) and von Mises equivalent stresses, and the factors of safety against yield, against a required factor.

$\sigma = K_t \cdot 32M/(\pi d^3)$; $\tau = K_{ts} \cdot 16T/(\pi d^3)$; $\tau_{max} = \sqrt{((\sigma/2)^2 + \tau^2)}$; $\sigma' = \sqrt{(\sigma^2 + 3\tau^2)}$; $n_T = (S_y/2)/\tau_{max}$; $n_{VM} = S_y/\sigma'$. The lower factor governs.

Ref: Shigley's Mechanical Engineering Design §5-4, §5-5 — Maximum-shear-stress and distortion-energy theories
Ref: Shigley's Mechanical Engineering Design §7-4 — Shaft design for stress

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
d	Shaft diameter	40	mm	drawing: DRG-SH-401 rev A	1 … 2000 mm
M	Bending moment (magnitude)	500	N*m	spec: LOAD-SCHEDULE-01	0 … 10000000 N*m
T	Torque (magnitude)	800	N*m	spec: LOAD-SCHEDULE-01	0 … 10000000 N*m
S_y	Yield strength	350	MPa	drawing: DRG-SH-401 rev A	50 … 3000 MPa
K_t	Bending stress-concentration factor (1 for none)	1		drawing: DRG-SH-401 rev A	1 … 5
K_ts	Torsional stress-concentration factor (1 for none)	1		drawing: DRG-SH-401 rev A	1 … 5
n_req	Required factor of safety	2		spec: SPEC-MECH-03 design criteria	1 … 10

3. Assumptions

A1. Solid round section, linear elastic; static loading. (Fatigue is mech.fatigue-miner's job; hollow shafts are not covered.)

4. Working

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[8] sigma = K_t * 32 * M / (pi * d^3)                    Shigley §7-4
= 1 × 32 × 500 N·m / (π × (40 mm)^3)
= 79.577 MPa
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Bending stress

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[9] tau = K_ts * 16 * T / (pi * d^3)                     Shigley §7-4
= 1 × 16 × 800 N·m / (π × (40 mm)^3)
= 63.662 MPa
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Torsional shear stress

[10] tau_max = sqrt((sigma / 2)^2 + tau^2) Shigley §5-4
= sqrt((79.577 MPa / 2)^2 + (63.662 MPa)^2)
= 75.073 MPa
Maximum shear stress (Tresca)
[11] sigma_vm = sqrt(sigma^2 + 3 * tau^2) Shigley §5-5
= sqrt((79.577 MPa)^2 + 3 × (63.662 MPa)^2)
= 135.98 MPa
von Mises stress
[12] n_T = (S_y / 2) / tau_max
= 350 MPa / 2 / 75.073 MPa
= 2.3311
Factor of safety, maximum shear stress
[13] n_VM = S_y / sigma_vm
= 350 MPa / 135.98 MPa
= 2.5739
Factor of safety, distortion energy
[14] n_min = min(n_T, n_VM)
= min(2.3311, 2.5739)
= 2.3311
Governing factor of safety
[15] u = n_req / n_min
= 2 / 2.3311
= 0.85798
Required over governing factor of safety

5. Results

SYMBOL	VALUE	DESCRIPTION
sigma	79.577 MPa	Bending stress
tau	63.662 MPa	Torsional shear
tau_max	75.073 MPa	Maximum shear stress
sigma_vm	135.98 MPa	von Mises stress
n_T	2.3311	Factor of safety (Tresca)
n_VM	2.5739	Factor of safety (von Mises)
n_min	2.3311	Governing factor of safety

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Factor of safety against the required factor	u = 0.85798	u ≤ 1	0.858	Shigley §7-4

7. Applicability

No warnings.

8. Verification

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